

Module-03
Shallow Foundation

Rajdhani

DATE / /

* **Book referred-** Soil Mechanics & foundation Engg. By B.C. Punmia.

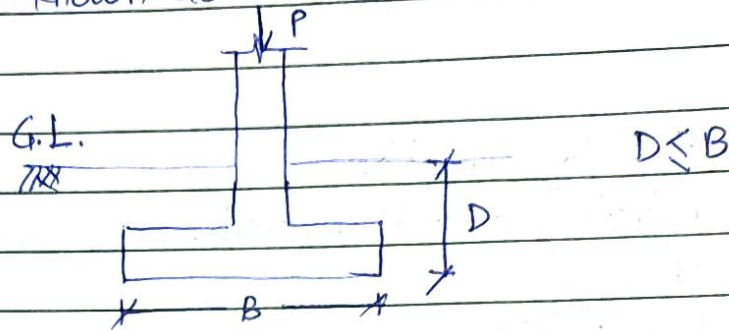
* **Contents-**

1. Introduction to Foundation.
2. Rigid analysis & elastic analysis.
3. Design of;
 - a) Isolated footing.
 - b) Combined footing.
 - c) Strap Footing.
 - d) Raft footing./ Mat footing.

* 1. **Introduction to Foundation**

- It is supporting member of structure which transfers load from building walls or columns to the earth.
- General requirements
 1. combine D.L., L.L., horizontal load like earthquake, w.L. can be resisted, carried and transmitted to the ground safely without any structural damage.
 2. It should be taken sufficient deep into the ground so that structure is not affected by ground movement such as swelling, shrinkage.
 3. Foundation base should be rigid so that differential settlement is minimised.
 4. Foundation should be so located that its performance should not be affected due to any unexpected future influence like earthquake & overloading.
 5. Foundation should resist attack from chemical in soil.
- shallow foundation-
When depth of footing is equal to or less than to its width

then it is known as shallow foundation.



This footing is suitable for moderate load and subsoil is having sufficient strength.

Types -

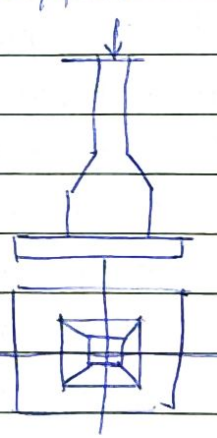
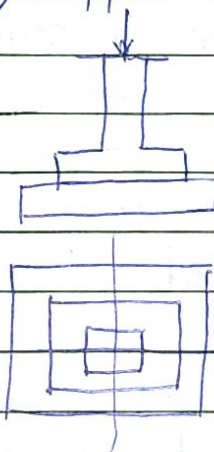
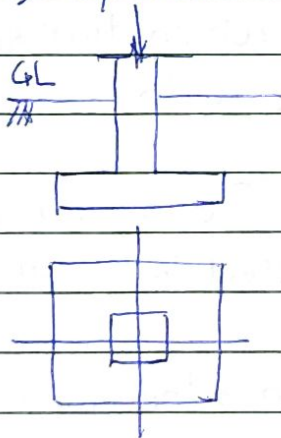
1. Isolated footing (For single column).

can be constructed in three ways

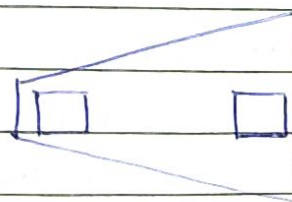
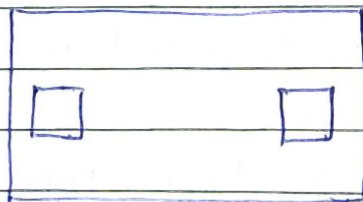
i) Spread

ii) Stepped

iii) Slopped

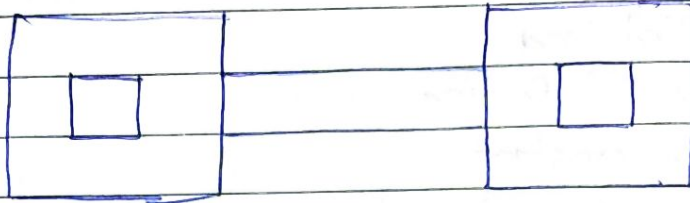


2. Combined Footing (supporting two adjacent columns)
Ex. Rectangular & trapezoidal.



It provides when column is lies at boundary of plot.

3. Strap Footing (Two isolated footing are connected by beam is called as strap footing)

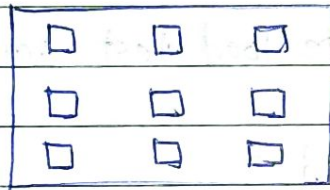


4. Strip OR continuous footing

Footing which is provided below wall OR row of column is called as strip OR continuous footing.

5. Mat/ Raft foundation

which supports many column together



* 2. Rigid analysis & elastic analysis

This point will be explained during design of mat foundation.

* 3. Design (shape, size, Depth, Reinforcement required)

1. Isolated footing

a) Shape -

- If load is axial - square footing
- If load is eccentric about both axis - square footing
- If load is eccentric about any one axis - Rectangular footing.

b) Size of footing

Size of footing is determined by using 'Housel's theory'.

As per Housels theory -

$$Q = A_m + P_n$$

Here; Q = Applied load

A = Area of footing

m = Area constant

P = Perimeter of footing.

n = Perimeter constant.

With the help of plate load test, performed on the ground where we have to construct footing, the values of m and n are determined & these values are further used to determine area & perimeter of footing by using above equation.

Example 01. The results of two plate load test for settlement of 25.4 mm are as given below.

Plate dia.	Load
0.305 m	31 kN
0.610 m	65 kN.

A square column foundation is to be design to carry a load of 800 kN. with an allowable settlement of 25.4 mm. Determine size of footing. using Housels method.

$$\begin{aligned}
 \rightarrow \quad A_1 &= \pi/4 \times d_1^2 & A_2 &= \pi/4 \times d_2^2 & P_1 &= \pi d_1 & P_2 &= \pi d_2 \\
 &= \pi/4 \times (0.305)^2 & &= \pi/4 \times (0.61)^2 & &= \pi \times 0.305 & &= \pi \times 0.61 \\
 &= 0.073 \text{ m}^2 & &= 0.292 \text{ m}^2 & &= 0.958 \text{ m} & &= 1.916 \text{ m}
 \end{aligned}$$

Use Housels eqⁿ

$$Q_1 = A_1 m + P_1 n$$

$$Q_2 = A_2 m + P_2 n$$

$$31 = 0.073 m + 0.292 n \quad \text{--- (1)}$$

$$65 = 0.958 m + 1.916 n \quad \text{--- (2)}$$

Solve eqⁿ ① & ②

We have $m = 20.55$, $n = 30.73$

Use these values to design square footing

$$Q = Am + Pn$$

$$800 = A(20.55) + P(30.73)$$

$$= B^2(20.55) + (4B)(30.73)$$

$$800 = 143.47$$

$$B = 3.93 \text{ m}$$

$$\text{Area of footing} = B^2 = 3.93 \times 3.93 = 15.44 \text{ m}^2$$

$$\text{Perimeter (P)} = 4B = 4 \times 3.93 = 15.72 \text{ m}$$

Example 2 The results of two plate load test on a given location are as follows

i) $D = 750 \text{ mm}$, settlement = 15 mm , ultimate load = 150 kN .

ii) $D = 300 \text{ mm}$, settlement = 15 mm , ultimate load = 50 kN .

Determine the ultimate load on a circular footing of 1.2 m dia. causing 15 mm settlement.

$$D_1 = 750 \text{ mm} = 0.75 \text{ m}$$

$$A_1 = \pi/4 \times 0.75^2 = 0.442 \text{ m}^2$$

$$P_1 = \pi D = \pi \times 0.75 = 2.356 \text{ m}$$

$$A_2 = \pi/4 \times 0.3^2 = 0.071 \text{ m}^2$$

$$P_2 = \pi D = \pi \times 0.3 = 0.942 \text{ m}$$

$$150 = 0.442 m + 0.256 n \quad \text{--- ①}$$

$$50 = 0.071 m + 0.942 n \quad \text{--- ②}$$

Solve eqⁿ ① & ②

$$m = 93.48, n = 46.13$$

$$\begin{aligned}
 Q &= A_m + P_n \\
 &= \pi/4 \times (1.2)^2 \times m + \pi \times 1.2 \times n \\
 &= \pi/4 (1.2)^2 \times 93.48 + \pi \times 1.2 \times 46.13
 \end{aligned}$$

$$Q_f = 279.54 \text{ KN}$$

Shear failure consideration & criteria -

$$\text{Area} = \text{Load on the column} + \left[\begin{array}{c} \text{self wt. of footing} \times \text{safe B.C of} \\ \text{footing} \quad \text{footing} \end{array} \right]$$

Self wt. of footing is considered as 10% of load on column.

③ Depth of footing

Vertical distance between the ground surface and base of footing is called as 'Depth of footing'.

It is max. of the following

$$① \quad D = q/\gamma \left[\frac{1 - \sin \phi}{1 + \sin \phi} \right]^2$$

where; q = SBC of soil

γ = sp. wt. of soil

ϕ = Angle of internal friction of soil.

$$② \quad D > 1.5$$

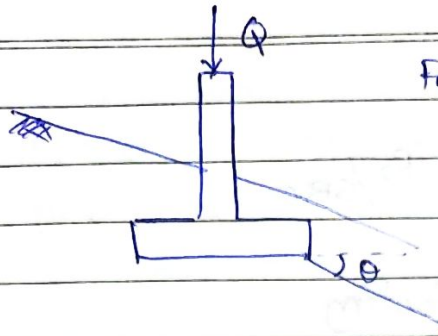
$D > \text{Root depth}$

$D > \text{Level of artificially increased temperature.}$

③ Frost depth

$D > 1.5 \text{ m (In cold areas)}$

④



For safe against erosion

$$\theta = 60^\circ \text{ - soil}$$

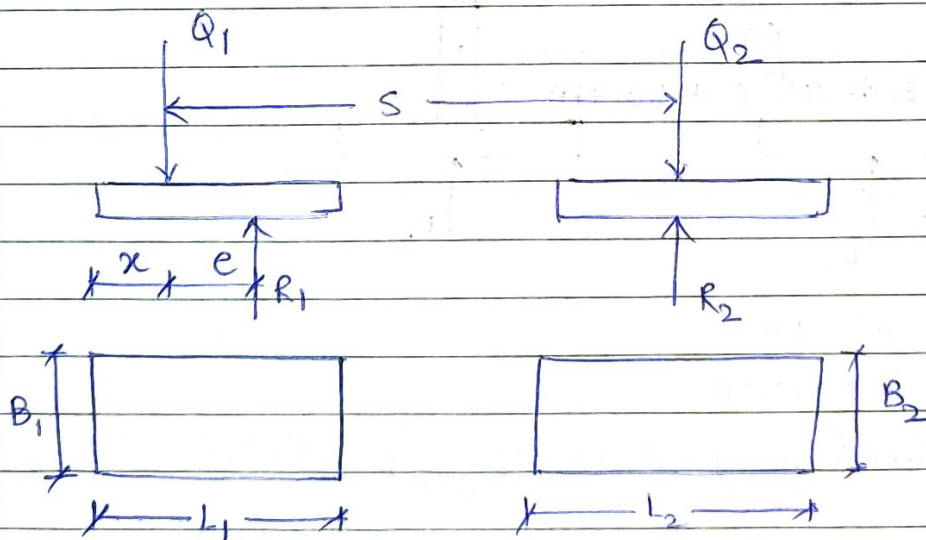
$$\theta = 30^\circ \text{ - Rock}$$

⑤ New footing should be below to the old footing.

2) Strap Footing

• Assumptions -

1. The strap footing, transfers the column load on the soil with equal & uniform soil pressure under both the footing.
2. The strap is pure flexure member.
3. It does not take directly soil reaction.
4. The soil below strap will be loose & the strap does not rest on soil & will not apply pressure.



• Steps -

1. Assume suitable value of e' (0.5 to 1.5 m)
2. Calculate soil reaction under each footing

$$R_1 = Q_1 \times \left(\frac{s}{s - e} \right), \quad R_2 = Q_1 + Q_2 - R_1$$

3. calculate area

$$A_1 = R_1 / \text{SBC}, \quad A_2 = R_2 / \text{SBC}$$

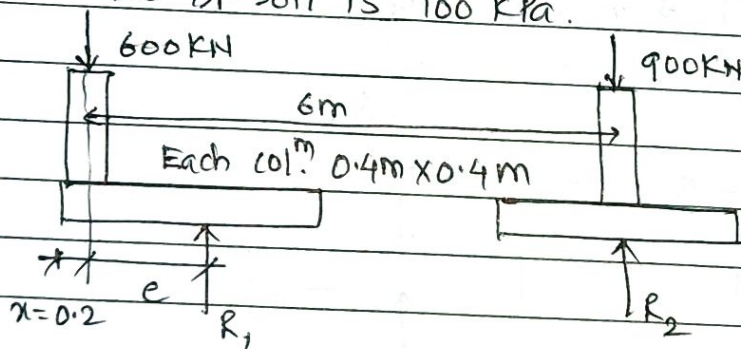
4. Calculate dimensions of footing

$$L_1 = 2(x + e)$$

$$B_1 = A_1 / L_1, \quad B_2 = B_1, \quad L_2 = A_2 / B_2$$

- Check that L & B ratio should not be too high. & $S > [0.5L_1 + 0.5L_2 + e]$
- If above conditions are not satisfy then revise the design by assuming some other value of e' .

Example: 1. Design a strap footing for the column as shown in fig. If SBC of soil is 100 kPa.

→ 1) Assume $e' = 1$ m

2) cal soil reaction

$$R_1 = Q_1 \times \left(\frac{S}{S - e} \right) = 600 \times \left(\frac{6}{6 - 1} \right), \quad R_1 = 720 \text{ kN}$$

$$R_2 = 600 + 900 - 720 = 780 \text{ kN}$$

3) calculate area

$$A_1 = \frac{R_1}{\text{SBC}} = \frac{720}{100} = 7.2 \text{ m}^2, \quad A_2 = \frac{R_2}{\text{SBC}} = \frac{780}{100} = 7.8 \text{ m}^2$$

4) Calculate dimensions of footing

$$L_1 = 2(x + e) = 2 \times (0.2 + 1) = 2.4 \text{ m}$$

$$B_1 = A_1 / L_1 = 7.2 / 2.4 = 3 \text{ m}$$

$$B_1 = B_2 \text{ i.e. } B_2 = 3 \text{ m}$$

$$L_2 = A_2 / B_2 = 7.8 / 3 = 2.6 \text{ m}$$

check i) $L_1 / B_1 = 2.4 / 3 = 0.8 < 5$

ii) $L_2 / B_2 = 2.6 / 3 = 0.86 < 5$

$$B = 0.5L_1 + 0.5L_2 + e$$

$$= (0.5 \times 2.4) + (0.5 \times 2.6) + 1$$

$$= 3.5 < 5 \text{ ——— OK.}$$

#3. Combined Footing

- Combined Footing is provided when,

i) Two columns are spaced too closely.

ii) Wall column is so close to the property line.

- Basic Principle -

Line of action of resultant load should coincide with the geometrical centre of footing and dimension of footing are chosen such that the allowable soil pressure is not exceeded.

- Types -

1) Rectangular (when two adjacent column carry equal load or less load at outer side)

Given,

a) Allowable soil pressure

b) Load carried by column

c) Distance of column from boundary.

Procedure -

1. Determine resultant column load $R = Q_1 + Q_2$
2. Determine area of footing $A_f = \frac{R}{q_a} = \frac{Q_1 + Q_2}{q_a}$
3. Locate position of resultant load w.r.t. external column
4. Determine position of centre of gravity of footing & length of footing.

$$\bar{x}_f = \bar{x}_L + W/2 \quad \text{--- ①}$$

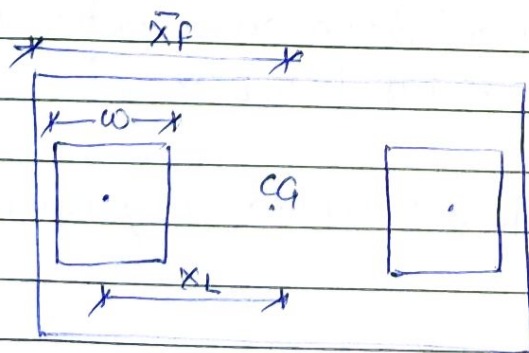
For rectangular footing

$$\bar{x}_f = L/2 \quad \text{--- ②}$$

By ① & ② Find 'L'

5. By using area & 'L', Find out width 'B'
6. Draw Shear force and BMD.

Example: 1. Design a Footing to support two column at center to center spacing of 6m carrying load of 3000 kN and 4500 kN. Lighter column is near the boundary 0.3m from edge of column. Assume column size 0.5 x 0.5 m and SBC is 350 kPa.



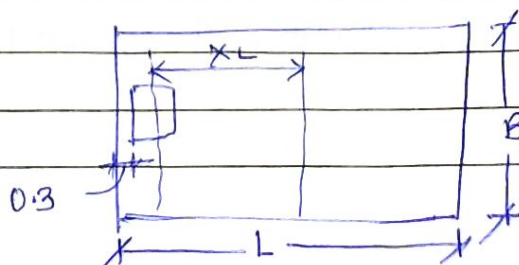
$$\begin{aligned} \text{① } R &= Q_1 + Q_2 \\ &= 3000 + 4500 = 7500 \text{ kN} \end{aligned}$$

$$\text{② } A_f = \frac{R}{q_a} = \frac{7500}{350} = 21.42 \text{ m}^2$$

$$\begin{aligned} \text{③ Position of resultant} \\ (7500 \times x_L) + (4500 \times 6) + 0 &= 0 \end{aligned}$$

$$x_L = 36 \text{ m}$$

$$\text{④ Position of CG of footing}$$



$$\bar{x}_f = \bar{x}_L + \frac{w}{2} + 0.3$$

$$= 3.6 + \frac{0.5}{2} + 0.3$$

$$\bar{x}_f = 4.15 \text{ m} \quad \text{--- (1)}$$

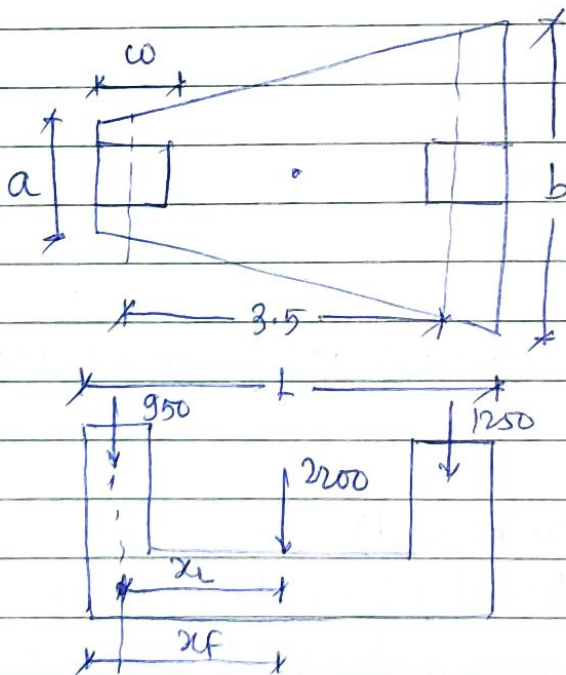
$$\bar{x}_f = \frac{L}{2} \quad \text{--- (2)}$$

From (1) and (2)

$$L = 8.3 \text{ m}$$

$$\text{width } B = A/L = \frac{21.4}{8.3} = 2.57 \text{ m}$$

Example: 2 Design combined footing (Trapezoidal) to support two columns carrying loads of 1250 kN and 950 kN resp. If the spacing between two columns is 3.5 m and size of each column is 0.3 m x 0.3 m and columns are flushed with the footing boundary on either side. SBC = 60 kPa



$$\textcircled{1} R = Q_1 + Q_2$$

$$= 1250 + 950$$

$$= 2200$$

$$\textcircled{2} \text{ Position of resultant,}$$

$$(2200 \times x_L) + (1250 \times 3.5) + 0 = 0$$

$$x_L = 2 \text{ m}$$

$$x_f = \frac{2 + 0.3}{2} = 2.15 \text{ m} \quad \text{--- (1)}$$

$\textcircled{3}$ calculate

$$A = \frac{a+b}{2} \times L$$

$$A = \frac{2200}{60} = 36.67 \text{ m}^2$$

$$\frac{a+b}{2} \times L = 36.67 \quad \text{--- (2)}$$

④ Position of CG from left end,

$$\bar{x}_f = \left(\frac{2a+b}{a+b} \right) \frac{L}{3} \quad \text{--- (3)}$$

$$2.15 = \frac{2a+b}{a+b} \times \frac{L}{3} \quad \text{--- (4)}$$

$$36.67 = \frac{a+b}{2} \times 3.8 \quad (L = 3.8) \text{ from eq. (2)}$$

$$\boxed{a+b = 19.3 \text{ m}} \quad \text{--- (5)}$$

From eq. (4)

$$\frac{3 \times 2.15}{3.8} = \frac{9}{a+b} + \frac{a+b}{a+b}$$

$$1.7 = \frac{9}{a+b} + 1$$

$$1.7 - 1 = \frac{9}{a+b}$$

$$-0.3a + 0.7b = 0 \quad \text{--- (6)}$$

From (5) and (6)

$$a = 13.51 \quad \& \quad b = 5.79 \text{ m}$$

#4. Mat Footing:

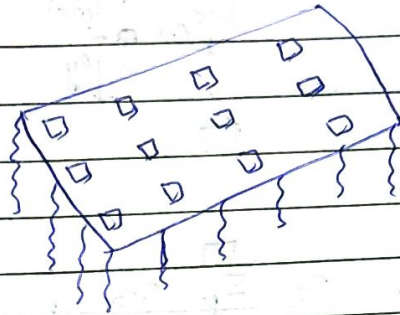
There are two approaches in design of mat or raft footing.

- 1) Elastic method (Flexible method)
- 2) Rigid method (conventional / IS code method)

The mat footing may be used where the base soil has low bearing capacity and column loads are so large such that 50% of the area is covered by conventional spread footings.

1. Elastic method -

It is assumed that the soil behave like an infinite number of individual springs not affected by adjacent springs.



The elastic constant of spring = The modulus of subgrade reaction of soil. Spring will resist both tension and compression. This concept is given by Winkler. so this method is also called as Winkler approach.

$$F = K \times \delta$$

Force in 'N' = Spring constant $\times$ amount of settlement or extension

2. Rigid method / Conventional method / IS-code method.

- This method can be used when settlement is small.
- when raft satisfies the condition ' $\lambda L < 1.75$ ' where, λ is the characteristic coefficient and 'L' is the common spacing.
- The characteristic coefficient is calculated by,

$$\lambda = \left[\frac{KB}{4E_c I} \right]^{0.25}$$

where; K = Modulus of subgrade reaction corrected for width B (kN/m³)

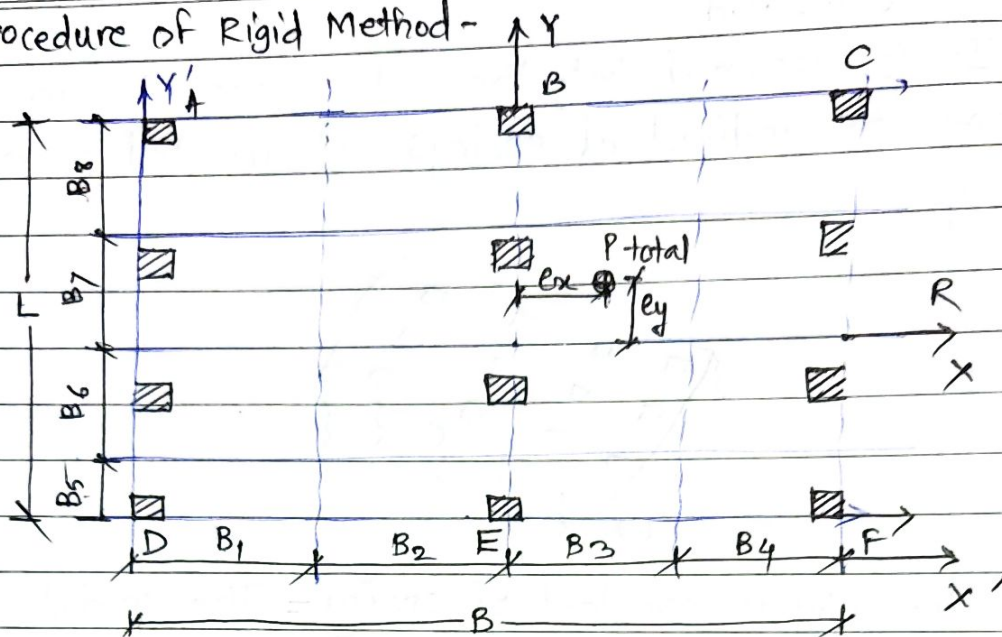
B = width of raft

E_c = Modulus of elasticity of concrete

$$= 5\sqrt{f_{ck}} \times 10^6 \text{ kPa}$$

I = Moment of inertia of raft in m⁴.

Procedure of Rigid Method -



- Let us consider a raft of dimension ($B \times L$) as shown above.
- $P_1, P_2, P_3 \dots$ Loads on the columns.
 R : Resultant column load
 I_x : MI of raft about X-axis = $BL^3/12$
 I_y : MI of raft about Y-axis = $LB^3/12$
 M_x = Moment about x axis
 $= R e_y + M_x (\text{Internal load})$
 M_y = Moment about y axis
 $= R e_x + M_y (\text{Lateral load})$
 e_x : Eccentricity of resultant load about centroid of footing about y axis.
 e_y : Eccentricity of resultant load about centroid of footing about x-axis.
- Choose reference axis in such way that entire raft lies in the first co-ordinate.
- Determine position of the resultant column load with respect to these axis.

$$x' = \frac{P_1 x_1 + P_2 x_2 + P_3 x_3 + \dots}{R} \text{ and } y' = \frac{P_1 y_1 + P_2 y_2 + P_3 y_3 + \dots}{R}$$

5. Determine eccentricity with respect to centre of footing.

$$e_x = x' - \frac{B}{2} \text{ and } e_y = y' - \frac{L}{2}$$

6. Divide the raft into number of strips parallel to y-axis (B_1, B_2, B_3) as well as parallel to x-axis (B_5, B_6 and B_7)

7. Draw SFD and BMD for all such strips.

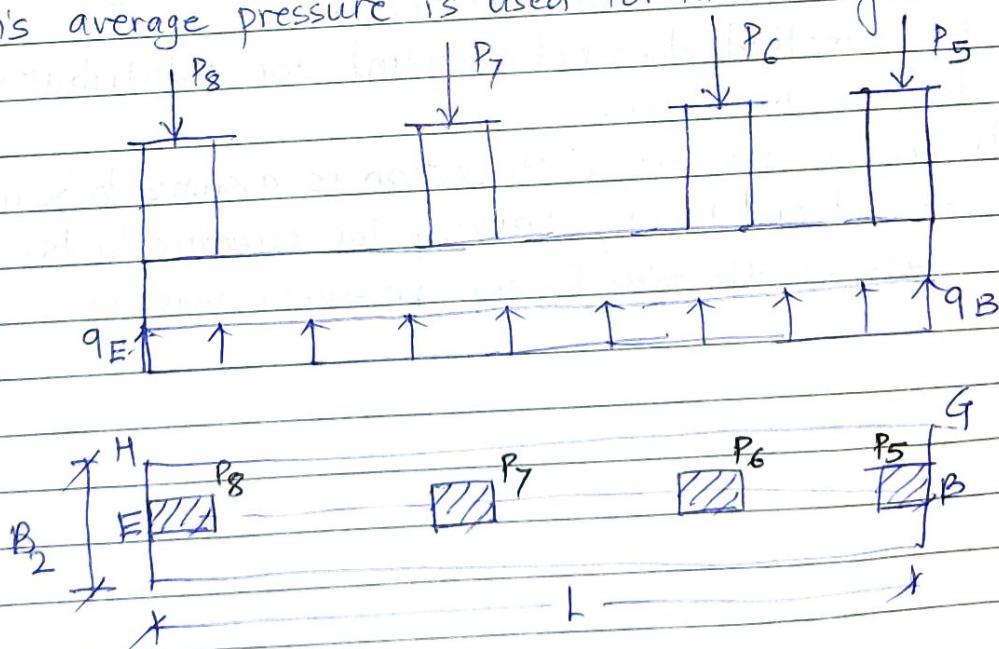
8. Sample calculation for one such strip GBIJEH. Soil pressure along the centre of this strip is assumed to be constant along the width of strip. Pressure at point B and E can be calculated by using the equation.

$$q = \frac{R}{A} \pm \frac{M_y}{I_y} x \pm \frac{M_x}{I_x} y$$

calculate average pressure on the strip:

$$q_{av} = \frac{q_B + q_E}{2}$$

This average pressure is used for the analysis of strip.



9. The strip has to satisfy equilibrium condition

Total downward load

$$= Q_D = P_5 + P_6 + P_7 + P_8$$

Total upward load

$$= Q_U = q_{av} (L B_2)$$

For equilibrium $Q_D = Q_U$

Average load on strip $q_{av} = \frac{Q_U + Q_D}{2}$

Modified soil pressure on per unit width $= w_m = \frac{q_{av}}{L}$

* Assumptions made in Rigid method.

- The footing/mat is infinitely rigid.
- The soil pressure is distributed in a straight line or plane surface.

* Limitations of Rigid method.

- In reality no foundation is rigid. some flexibility will be there.
- Portion of mat below columns are bearing walls settle more than the portion with less load.
- Rigid method does not account for redistribution of bearing pressure.
- In general, pressure distribution is assumed to be uniform for axial load and linearly varying for eccentric loads.
- In case of flexible footing, pressure is uniform